Supplementary Material

A Keypoint Representation

We use a 61-keypoint 3D pose representation for each frame. Each keypoint contains (x, y, z) coordinates, giving $D = 61 \times 3 = 183$ pose values per frame. The layout is shown in Fig. A1.

- **Body keypoints (0–18):** upper torso, arms, hips, and facial landmarks. This includes shoulders, elbows, wrists, hips, the neck/head base, and facial points around the eyes, ears, and mouth.
- **Right-hand keypoints (19–39):** one wrist keypoint and five four-joint finger chains.
- **Left-hand keypoints (40–60):** one wrist keypoint and five four-joint finger chains, following the same layout as the right hand.

B Gloss-Motion Dictionary Construction

We construct the gloss-motion dictionary from the training split. The dictionary stores multiple real motion segments for each gloss, which are later used to initialize the retrieved sequence in SignRR. Since the datasets provide sentence-level gloss annotations but not frame-level gloss boundaries, we estimate segment boundaries using CorrNet+ [1] and the known gloss order.

Boundary estimation. Given a training sequence with glosses $G = (g_1, \ldots, g_N)$, we run the pretrained CorrNet+ recognizer to obtain a frame-level score $r_t(g)$ for each gloss g at frame t. We then use the ground-truth gloss order to estimate the start of each gloss. For gloss g_i, the start frame is selected as the first frame after the previous boundary where the CorrNet+ score of g_i becomes larger than the score of the previous gloss:

$$b_i = \min\{t > b_{i-1} \mid r_t(g_i) > r_t(g_{i-1})\}. \tag{A1}$$

This gives a weak temporal alignment without requiring frame-level boundary annotations. The segment for g_i is extracted from the interval between consecutive boundaries.

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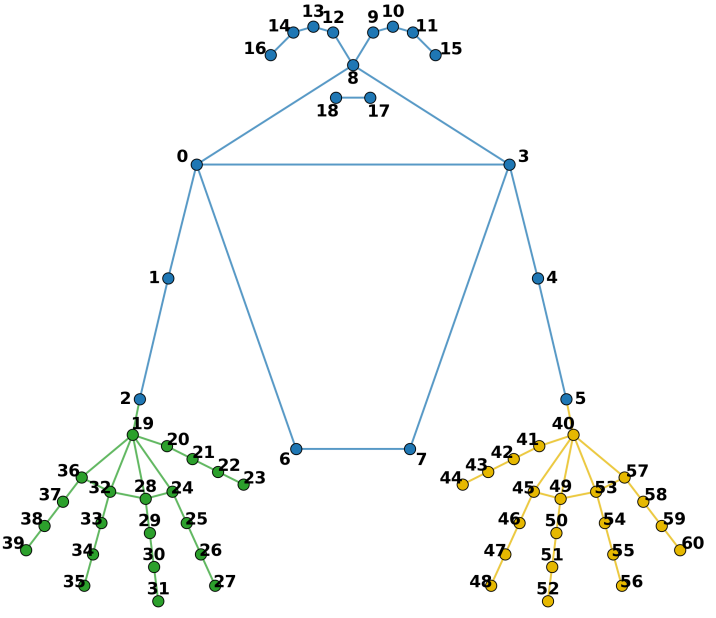


Figure A1: Keypoint layout. Blue nodes denote body keypoints, green nodes denote right-hand keypoints, and yellow nodes denote left-hand keypoints.

Candidate filtering. The raw aligned segments can contain noisy boundaries or abnormal durations. We therefore filter candidates independently for each gloss. Let $\ell(s)$ denote the duration of segment s, and let μ_g and σ_g be the mean and standard deviation of the segment durations for gloss g. A segment is discarded if its duration is too far from the typical duration of that gloss:

$$|\ell(s) - \mu_g| > \tau_\ell \sigma_g. \tag{A2}$$

This removes very short or very long segments that are likely to come from poor alignment.

Candidate scoring. After filtering, each remaining segment is scored using motion smoothness and duration regularity. The smoothness term favors segments with stable temporal motion. For a segment $s = (\mathbf{x}_1, \ldots, \mathbf{x}_\ell)$, we measure smoothness using the average frame-to-frame acceleration:

$$e_{\text{sm}}(s) = \frac{1}{\ell - 2} \sum_{t=2}^{\ell-1} \|\mathbf{x}_{t+1} - 2\mathbf{x}_t + \mathbf{x}_{t-1}\|_2 . \tag{A3}$$

Lower values indicate smoother motion. We convert this error into a normalized score:

$$q_{\text{sm}}(s) = \exp\left(-e_{\text{sm}}(s)\right). \tag{A4}$$

The duration term favors segments whose length is close to the typical duration of the same gloss:

$$q_{\text{dur}}(s) = \exp\left(-\frac{|\ell(s) - \mu_g|}{\sigma_g + \varepsilon}\right). \tag{A5}$$

Algorithm 1 Gloss-motion dictionary construction

Require: Training sequences with sentence-level gloss annotations; pretrained CorrNet+ recognizer
Ensure: Gloss-motion dictionary $\mathcal{D}$

```
Initialize an empty candidate set for each gloss
for each training sequence do
    Run CorrNet+ to obtain frame-level gloss scores
    Estimate gloss boundaries using Eq. A1
    for each aligned gloss occurrence do
        Extract the corresponding motion segment
        Add the segment to the candidate set of that gloss
    end for
end for
for each gloss g do
    Compute duration statistics μ_g and σ_g
    Remove abnormal-duration segments using Eq. A2
    Score each remaining segment using Eq. A6
    Keep the top S candidates
    Store each retained segment and its boundary descriptor in D(g)
end for
return D
```

The final candidate score combines both terms:

$$q(s) = \lambda_{\text{sm}} q_{\text{sm}}(s) + \lambda_{\text{dur}} q_{\text{dur}}(s), \tag{A6}$$

where λ_{sm} and λ_{dur} control the relative weights of motion smoothness and duration regularity. For each gloss, we keep the top S candidates according to $q(s)$.

Boundary descriptors. For each retained segment, we also store a boundary descriptor used by the dynamic-programming selector during inference. The descriptor contains the first and last poses of the segment, together with their local velocities:

$$\phi(s) = [\mathbf{x}_1,\ \mathbf{x}_\ell,\ \mathbf{x}_2 - \mathbf{x}_1,\ \mathbf{x}_\ell - \mathbf{x}_{\ell-1}]. \tag{A7}$$

This descriptor allows the selector to compare the compatibility of adjacent retrieved segments without using the full segment sequence.

The resulting dictionary provides real motion candidates for each gloss. The filtering step reduces noisy alignments, while the scoring step favors segments that are both temporally smooth and close to the typical duration of the corresponding gloss.

C Dynamic Programming Segment Selection

At inference time, SignRR receives a gloss sequence and selects one motion segment per gloss from the dictionary $\mathcal{D}$. Selecting segments only by their individual quality can produce visible jumps at segment boundaries, while selecting only by boundary smoothness can ignore the quality of the retrieved signs. We therefore optimize both terms jointly using dynamic programming over the candidate trellis defined by $\mathcal{D}$.

Out-of-dictionary glosses. A gloss in the test input may be absent from $\mathcal{D}$ because it was filtered out during dictionary construction (Sec. B) or because of minor spelling variants. We replace each such gloss with the closest dictionary entry $g^\star$ under Levenshtein edit distance:

$$g^\star = \arg\min_{g' \in \mathcal{V}_\mathcal{D}} \mathrm{Lev}(g, g'), \tag{A8}$$

where $\mathcal{V}_\mathcal{D}$ is the set of glosses present in the dictionary. This ensures that each gloss position has at least one candidate segment. In the two evaluated datasets, unseen gloss types do not often occur. PHOENIX14T contains 19 in the development split and 22 in the test split, while CSL-Daily contains none.

Boundary cost. The boundary cost measures the discontinuity between two consecutive segments. Given a previous segment s' and a current segment s, we compare the end descriptor of s' with the start descriptor of s:

$$d_{\mathrm{bnd}}(s', s) = \left\| \phi_{\mathrm{end}}(s') - \phi_{\mathrm{start}}(s) \right\|_2. \tag{A9}$$

The descriptors include both pose and local velocity, so this cost penalizes spatial jumps and abrupt motion changes at the seam. Since d_{bnd} and the segment quality score $q(s) \in [0,1]$ have different scales, we normalize the boundary cost by a dataset-level constant $\tilde{d}$. We compute $\tilde{d}$ once as the median boundary cost over randomly sampled pairs of dictionary segments, making the normalization robust to outliers.

Dynamic-programming recurrence. Let $\mathcal{S}_i = \mathcal{D}(g_i)$ be the candidate set for gloss g_i, and let $s_{i,j} \in \mathcal{S}_i$ denote its j-th candidate. We define $C_i(j)$ as the minimum cumulative cost of any path ending at $s_{i,j}$. The first position is initialized using only the segment quality cost:

$$C_1(j) = \beta\big(1 - q(s_{1,j})\big). \tag{A10}$$

For $i = 2, \ldots, N$, the recurrence is

$$C_i(j) = \min_k \left[C_{i-1}(k) + \alpha \frac{d_{\mathrm{bnd}}(s_{i-1,k}, s_{i,j})}{\tilde{d}} \right] + \beta\big(1 - q(s_{i,j})\big), \tag{A11}$$

where α and β control the trade-off between transition smoothness and segment quality. We store the minimizing predecessor in a back-pointer table $B_i(j)$ and use $\alpha = 1.0$ and $\beta = 0.5$ in all experiments.

Backtracking. After the forward pass, we start from the lowest-cost candidate at the final gloss:

$$j_N^\star = \arg\min_j C_N(j). \tag{A12}$$

We then recover the full path by following the back-pointers:

$$j_i^\star = B_{i+1}(j_{i+1}^\star), \qquad i = N-1, \ldots, 1. \tag{A13}$$

The selected path is $\pi^\star = (\pi_1^\star, \ldots, \pi_N^\star)$, where $\pi_i^\star = s_{i,j_i^\star}$. These segments are concatenated in order and passed to the RVQ-VAE refinement model as the initial motion.

With at most S candidates per gloss and a gloss sequence of length N, the algorithm runs in $O(NS^2)$ time. The selected path balances segment quality and boundary continuity, giving the refinement model a smoother retrieved initialization.

Algorithm 2 Dynamic programming segment selection

Require: Gloss sequence $G = (g_1, \ldots, g_N)$; dictionary $\mathcal{D}$; weights α, β; normalizer $\tilde{d}$
Ensure: Selected path $\pi^\star = (\pi_1^\star, \ldots, \pi_N^\star)$

```
Resolve out-of-dictionary glosses in G using Eq. A8
Let S_i = D(g_i) for i = 1, ..., N
for j = 1, ..., |S_1| do
    C_1(j) ← β(1 − q(s_{1,j}))
end for
for i = 2, ..., N do
    for j = 1, ..., |S_i| do
        Compute C_i(j) using Eq. A11
        Store the minimizing predecessor in B_i(j)
    end for
end for
j*_N ← arg min_j C_N(j)
for i = N − 1, ..., 1 do
    j*_i ← B_{i+1}(j*_{i+1})
end for
π*_i ← s_{i,j*_i} for i = 1, ..., N
return π*
```

D Implementation Details

This section summarizes the implementation details used in our experiments.

Refinement model. The refinement model uses part-specific encoders for the body, right hand, and left hand. Each encoder has two stride-2 downsampling stages with dilated ResNet-1D blocks and hidden width 512. Each part uses $L = 3$ residual codebooks. The body stream uses codebooks with 256 entries, while each hand stream uses codebooks with 384 entries. The shared decoder mirrors the encoder structure and outputs the $D = 183$ pose dimensions.

Gloss encoder and length prediction. The gloss encoder is a 2-layer Transformer with 4 attention heads, model dimension 128, and dropout 0.1. Its mean-pooled output is projected to 512 dimensions and passed to a 2-layer ReLU MLP with hidden width 256 to predict the length offset Δ. We use $T_{\max} = 300$ for PHOENIX14T, $T_{\max} = 352$ for CSL-Daily, and $\delta_{\text{scale}} = 100$. At inference, the predicted target length is clamped to $[16, T_{\max}]$.

Training objective. For the training objective in the main paper, we use $\lambda_v = 0.1$, $\lambda_c = 0.02$, and $\lambda_\ell = 0.01$. In the part-weighted smooth-ℓ_1 reconstruction loss, the body weight is 1.0, and each hand weight is 2.0.

Dictionary configuration. For each gloss, we keep at most $S = 25$ representative segments. Candidates are selected using the quality score described in Sec. B, with diversity-based subsampling when more than S candidates are available. The dynamic-programming normalizer $\tilde{d}$ is estimated once per dataset from randomly sampled inter-gloss segment pairs.

After filtering, the dictionaries contain 9,311 segments for PHOENIX14T and 32,500 segments for CSL-Daily, with an average of 8.61 and 16.26 candidates per gloss, respectively.

E Synthetic Motion Generation

For our refinement-without-retrieval analysis, we use a synthetic motion to evaluate whether the refinement model can operate without real retrieved sign segments. We synthesized continuous hand displacement trajectories by adding mathematically defined motion curves to a randomly selected reference frame. We used three trajectory types to approximate simple hand motion patterns: linear (Eq. A14), sinusoidal (Eq. A16), and quadratic (Eq. A15). Each trajectory used a fixed amplitude chosen empirically, while the orientation and frequency were randomized to increase variation across samples.

Let n denote the number of frames in the synthetic sequence, and let $s \sim \mathcal{U}(0.005, 0.02)$ be the step size sampled uniformly at random. The displacement domain is defined as an evenly spaced sequence of n points. Each trajectory returns a sequence of 2D displacement vectors $\{(x_i, y_i)\}_{i=0}^{n-1}$, where $x_i = i \cdot s$ and $i = 0, 1, \ldots, n-1$. The three trajectories are defined as follows:

$$(x_i,\ y_i) = \begin{cases} (0,\ i \cdot s) & \text{if vertical,} \\ (i \cdot s,\ 0) & \text{if horizontal.} \end{cases} \tag{A14}$$

$$x_i = i \cdot s, \qquad y_i = 12 x_i^2 \tag{A15}$$

$$x_i = i \cdot s, \qquad y_i = 0.1 \sin(25 x_i) \tag{A16}$$

The resulting displacement vectors are added to the X and Y components of the keypoints from the randomly selected reference frame $\mathbf{p}^{(\text{ref})}$, producing the displaced keypoint positions $\mathbf{p}_i$:

$$\mathbf{p}_i = \mathbf{p}^{(\text{ref})} + \mathbf{d}_i, \quad \mathbf{d}_i = (x_i,\ y_i) \in \mathbb{R}^2. \tag{A17}$$

The trajectories were applied to 46 selected keypoints per frame, since these keypoints carry most of the movement during signing. The selected keypoints cover the wrists, palms, and finger joints of both hands. Specifically, the right hand includes keypoints 1–2 and 19–39, while the left hand includes keypoints 4–5 and 40–60.

To produce temporally smooth sequences, adjacent segments were joined using linear boundary interpolation (Eq. A18). Given two consecutive segments $S^{(k)}$ and $S^{(k+1)}$, let $\mathbf{p}_{\text{last}}^{(k)}$ and $\mathbf{p}_{\text{first}}^{(k+1)}$ denote their boundary frames. A fixed number of $T = 10$ interpolated frames is inserted between them:

$$\mathbf{p}_t = (1 - \alpha_t)\, \mathbf{p}_{\text{last}}^{(k)} + \alpha_t\, \mathbf{p}_{\text{first}}^{(k+1)}, \quad t = 1, \ldots, T, \qquad \alpha_t = \frac{t}{T+1}. \tag{A18}$$

Since $\alpha_t \in (0, 1)$, the boundary frames are not duplicated, and the transitions remain smooth. All segments are then concatenated to produce synthetic sequences with lengths comparable to the original sequences, allowing a variation of ± 20 frames per video.

We then applied a filtering step to remove physically implausible frames based on two anatomical constraints. A frame is marked as invalid if either of the following rules is violated:

Table A1: Robustness to dictionary segmentation on PHOENIX14T (TEST). Each setting rebuilds the dictionary and retrains SignRR using the corresponding boundaries. **Bold** marks the best value in each column.

Dictionary	WER↓	BLEU-1↑	BLEU-4↑	ROUGE↑	MPJPE↓
SignRR (proposed)	80.98	28.78	10.96	28.15	53.70
Jitter ±4	81.43	28.28	11.28	27.90	53.47
Jitter ±8	82.08	27.57	10.65	27.21	54.05
Jitter ±16	82.88	27.00	10.54	26.53	54.41
Uniform (no recognizer)	81.40	27.26	10.58	27.12	53.55

- **Hand height:** no keypoint belonging to either hand may exceed a vertical ceiling defined by keypoint 9 plus an offset of 0.6 units. This prevents the hands from being placed far above the head.
- **Arm length:** the elbow keypoints, keypoints 1 and 5, must remain within 20% of the vertical distance between keypoints 0 and 6, which corresponds to the torso length. This prevents physically implausible arm extensions.

After filtering, the remaining valid frames are resampled back to the original sequence length using linear interpolation, following the same procedure as Eq. A18.

The resulting synthetic sequences approximate simple continuous hand motion patterns while remaining semantically neutral; that is, they do not correspond to any actual sign or meaningful gesture. This provides smooth motion inputs that lack sign-specific articulation, allowing us to test whether real retrieved motion is necessary for effective refinement.

F Robustness to Dictionary Segmentation

We evaluate the robustness of SignRR to variations in the gloss boundaries used to construct the dictionary. For each interior boundary, we independently sample an offset from the discrete uniform distribution over $\{-k,\dots,k\}$ frames, with $k \in \{4,8,16\}$, and shift the boundary by that offset. We constrain the perturbed boundaries to preserve gloss order and a minimum segment length of three frames, and keeping the first and last sequence boundaries fixed. For each setting, we rebuild the dictionary from the perturbed segments and retrain SignRR using the same configuration as in the main experiments.

We also consider a recognizer-free baseline in which a sequence of length L containing N glosses is uniformly divided into N segments without using CorrNet+ predictions. The same dictionary construction and training procedure is then applied. Table A1 shows that performance remains stable under perturbations and decreases gradually as the boundary variation increases. At ±16 frames, BLEU-4 decreases by only 0.42. Uniform segmentation still obtains BLEU-4 10.58, showing that CorrNet+-based boundaries are helpful, while SignRR remains effective with a much coarser segmentation.

G Inference Time

We evaluate inference time on the PHOENIX14T test set with batch size 1 on a single NVIDIA A100 GPU. Dynamic-programming selection runs on the CPU, while refinement is measured as one forward pass of the full model.

Table A2: Per-sequence inference time on PHOENIX14T (TEST) with batch size 1 on a single NVIDIA A100 GPU.

Method	ms/seq.	Params (M)
SignRR: DP selection	54.6	–
SignRR: assembly	0.1	–
SignRR: refinement	33.1	38.5
SignRR: total	**87.8**	**38.5**
Sign-IDD [2]	85.0	16.8
G2P-DDM [3]	2237	106.3

Table A2 reports the inference time for each component and the parameter counts. Sign-IDD and G2P-DDM are measured using their released implementations on the same GPU with 10 DDIM and 100 diffusion steps, respectively. SignRR requires 87.8,ms per sequence, with 54.6,ms for segment selection and 33.1,ms for refinement. Its total inference time is comparable to Sign-IDD and about 25$\times$ faster than G2P-DDM.

References

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[3] Pan Xie, Qipeng Zhang, Peng Taiying, Hao Tang, Yao Du, and Zexian Li. G2p-ddm: Generating sign pose sequence from gloss sequence with discrete diffusion model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 6234–6242, 2024.